```
% ================================
% FLANGE DESIGN CALCULATION SCRIPT
% ME 329 Piping Flange System Design
% Authors: Gabriel Choi, Ryan Trevena
% ================================

% ---------- Given/selected values ----------
P = 400;                          % psi, Maximum Expected Operating Pressure
r_i = (12.75 - (2*0.156)) / 2;    % inch, Internal radius of the pipe (based on pipe wall thickness) from engineers edge standard pipe schedules 12 in nominal
r_o = 12.75 / 2;                  % inch, Outer radius from standard pipe schedule
E = 1.0;                          % Weld efficiency (assumed perfect)
T_su = 0.89;                      % Temperature knockdown factor for ultimate strength
T_sy = 0.73;                      % Temperature knockdown factor for yield strength
```

---------- Step 1: Wall Thickness Calculation ----------

```
thickness = r_o - r_i;            % Actual wall thickness of the pipe

% Material Properties (Inconel 625 at 600°F)
% Returns temperature-dependent yield and ultimate strengths based on thickness
function [Stu, Sty] = inconel625_properties(thickness)
    if thickness <= 0.062
        Stu = 119e3; Sty = 56e3;
    elseif thickness <= 0.109
        Stu = 119e3; Sty = 55e3;
    elseif thickness <= 0.140
        Stu = 119e3; Sty = 54e3;
    elseif thickness <= 0.187
        Stu = 118e3; Sty = 53e3;
    elseif thickness <= 0.250
        Stu = 119e3; Sty = 59e3;
    elseif thickness <= 1.000
        Stu = 120e3; Sty = 60e3;
    else
        error('Thickness out of range for Inconel 625.');
    end
end

[Stu, Sty] = inconel625_properties(thickness);
Stu_T = Stu * T_su;               % Apply temperature knockdown
Sty_T = Sty * T_sy;

% Calculate hoop, radial, and axial stresses
F_tmax = P * (((r_o^2) + (r_i^2)) / ((r_o^2) - (r_i^2)));
```

```
F_rmax = -P;
F_lmax = (P * r_i^2) / ((r_o^2) - (r_i^2));

% Von Mises equivalent stress
F_total = sqrt(((F_tmax - F_lmax)^2 + (F_lmax - F_rmax)^2 + (F_rmax -
F_tmax)^2) / 2);

% Safety factors based on yield and ultimate strength
SFy = Sty_T / F_total;
SFu = Stu_T / F_total;

fprintf('=== Wall Thickness Calculation ===\n');
fprintf('Required Wall Thickness: %.4f in\n', thickness);
fprintf('Safety Factor (Yield): %.2f\n', SFy);
fprintf('Safety Factor (Ultimate): %.2f\n', SFu);
```

```
=== Wall Thickness Calculation ===
Required Wall Thickness: 0.1560 in
Safety Factor (Yield): 2.70
Safety Factor (Ultimate): 7.33
```

---------- Step 2: Seal Area and Required Clamp Load (For a Seal) ----------

```
D_Oseal = 16.025;                          % Outer seal diameter (in)
D_Iseal = 15.475;                          % Inner seal diameter (in)
T_seal = D_Oseal - D_Iseal;               % Seal width (in)
A_seal = pi * ((D_Oseal / 2)^2 );  % Seal area (in^2)

P_seal_required = 400;                     % Required sealing pressure (psi)
from seal spec sheet
W_seal_required = A_seal * P_seal_required;

fprintf('\n=== Seal Area and Required Clamp Load (Mechanical Seal) ===\n');
fprintf('Seal Area: %.2f in^2\n', A_seal);
fprintf('Required Sealing Pressure: %.2f psi\n', P_seal_required);
fprintf('Total Required Clamp Load (W): %.2f lb\n', W_seal_required);
```

```
=== Seal Area and Required Clamp Load (Mechanical Seal) ===
Seal Area: 201.69 in^2
Required Sealing Pressure: 400.00 psi
Total Required Clamp Load (W): 80676.30 lb
```

---------- Step 3: Flange Thickness and Diameter Calculation ----------

```
T = 1;                          % Initial flange thickness estimate (in)
w_flange = 3;                      % Flange widtch from pipe to edge
H = 1.5 * T;                     % Flange height estimate (in)
R_i = 12.75 / 2;                   % Inner flange radius (in)
```

```
W_total = W_seal_required;   % Total axial load on flange (seal + preload)
R_O = R_i+w_flange;                % Outer flange radius (in)
m = R_i / T;                     % Geometry ratios for Waters-Taylor formula
n = H / T;
k = (R_i + w_flange) / R_i;

% Waters-Taylor bending stress formula
numerator = (0.25 * sqrt(m) * ((k^2) * (1 + 8.55 * log(k)) - 1));
denominator = ((1.05 + 1.94 * (k^2)) * (sqrt(m) + 0.64 * n) + 0.53 * (n^3) *
(k^2 - 1));
sigma_bmax = (W_total / (T^2)) * (numerator / denominator);  % psi

[Stu_f, Sty_f] = inconel625_properties(T);
Stu_F = Stu_f * T_su;
Sty_F = Sty_f * T_sy;

SF_yield = Sty_F / sigma_bmax;
SF_ultimate = Stu_F / sigma_bmax;

fprintf('\n=== Flange Thickness and Bending Stress ===\n');
fprintf('Thickness: %.2f\n', T);
fprintf('Height: %.2f\n', H);
fprintf('Flange Outer Radius: %.2f\n', R_O);
fprintf('Maximum Bending Stress (incl. preload): %.2f psi\n', sigma_bmax);
fprintf('Safety Factor (Yield): %.2f\n', SF_yield);
fprintf('Safety Factor (Ultimate): %.2f\n', SF_ultimate);
```

```
=== Flange Thickness and Bending Stress ===
Thickness: 1.00
Height: 1.50
Flange Outer Radius: 9.38
Maximum Bending Stress (incl. preload): 20744.67 psi
Safety Factor (Yield): 2.11
Safety Factor (Ultimate): 5.15
```

----------Step 4.a Preload Calculation----------

```
n_bolts = 24;                              % Number of bolts
d_bolt = 0.75;                             % Bolt diameter (in)
At = 0.334;                              % Tensile stress area for 1/2"-13 UNC
(in^2)

% Bolt strength data at room temperature
Sp = 132e3; Sy = 155e3; Su = 180e3;
Sp_T = Sp * T_su; Sy_T = Sy * T_sy; Su_T = Su * T_su;

% Calculate default and required preload per bolt
Fi = 0.45 * Sp_T * At;                                % Default preload (50%
of proof strength)
Fi_required = W_seal_required / n_bolts;           % Minimum preload
required for seal
```

```
if Fi_required < Fi
    fprintf('\nNote: Preload per bolt increased above 25%% proof strength to
meet sealing requirement.\n');
end
```

Note: Preload per bolt increased above 25% proof strength to meet sealing requirement.

--------- Step 4.b: Fastener Analysis ---------

```
Fy = Sy_T * At;                          % Yield capacity of bolt
Fu = Su_T * At;                          % Ultimate capacity of bolt

W_total_preload = Fi * n_bolts;                        % Total preload load
applied
W_per_bolt = W_total_preload / n_bolts;

SFy_no_preload = Fy / W_per_bolt;
SFu_no_preload = Fu / W_per_bolt;

% Joint stiffness model parameters
E_b = 28e6;                              % psi, modulus of bolt
E_m = 26e6;                              % psi, modulus of member
L_b = 1.0;                               % in, bolt grip length
L_m = 1.0;                               % in, member compression length
A_m = 2 * At;                            % in^2, estimated compression area
under head

k_b = At * E_b / L_b;
k_m = A_m * E_m / L_m;
C = k_b / (k_b + k_m);                   % Load fraction carried by bolt

Fb = Fi + C * W_per_bolt;               % Total bolt load = preload +
external load share

SFy_with_preload = Fy / Fb;
SFu_with_preload = Fu / Fb;
%------Torque Calc-----%

K = 0.18;                                % Torque coefficient (lubricated)
Torque = K * Fi * d_bolt;                % Torque required for preload


%--- Flange separation check---%
W_separation = C * W_total;                % Load trying to separate the joint
Fi_total = Fi * n_bolts;             % Total clamp load applied by bolts
SF_sep = Fi_total / W_separation;    % Safety factor against flange separation

% === Socket/Wrench Clearance Checker ===

n_bolts = 24;                     % Number of bolts
BCD = 17;                         % Given Bolt Circle Diameter (in)
```

```
R_bolt_circle = BCD / 2;          % Radius of bolt circle
socket_OD = 1.375;                % Socket outer diameter (in)

% CAD-measured radial clearance
radial_clearance = 1.125;                     % in (from CAD)
R_outer_flange = R_bolt_circle + radial_clearance;
OD_flange = 2 * R_outer_flange;

% Bolt-to-bolt spacing
angle = 360 / n_bolts;
spacing = 2 * R_bolt_circle * sind(angle / 2);   % in
actual_margin = spacing - socket_OD;
actual_radial_margin = radial_clearance - socket_OD/2;

fprintf('=== Socket Clearance Report ===\n');
fprintf('Bolt Circle Diameter: %.3f in\n', BCD);
fprintf('Flange Outer Diameter: %.3f in\n', OD_flange);
fprintf('Bolt-to-Bolt Spacing: %.3f in\n', spacing);
fprintf('Radial Clearance (Bolt to Wall): %.3f in\n', radial_clearance);
fprintf('---\n');
fprintf('Socket OD: %.3f in\n', socket_OD);
fprintf('Actual Margin Between Bolts: %.3f in\n', actual_margin);
fprintf('Actual Radial Margin to Wall: %.3f in\n', actual_radial_margin);

if actual_margin >= 0.2 && actual_radial_margin >= 0.2
    fprintf('□ PASS: Socket fits with sufficient margin (≥ 0.2 in).\n');
elseif actual_margin < 0.2 && actual_radial_margin >= 0.2
    fprintf('□ FAIL: Not enough space between bolts for socket.\n');
elseif actual_margin >= 0.2 && actual_radial_margin < 0.2
    fprintf('□ FAIL: Not enough radial space to flange wall.\n');
else
    fprintf('□ FAIL: Socket clearance failed in both directions.\n');
end


% === O-Ring Gland Dimension Calculator (Axial Face Seal) ===

d_cross = 0.275;           % O-ring cross-sectional diameter (in)
compression_target = 0.80; % Compression as fraction (e.g., 80%)
fill_target = 0.85;         % Groove fill percentage

% Axial face seal groove dimensions (general standards)
groove_depth = d_cross * compression_target;  % Groove depth = 80% of cross-
section
groove_width = d_cross / fill_target;         % Groove width = enough to
limit fill
land_width = 0.050;                            % Land between groove ID and
OD (typical)

% Estimate gland ID and OD for an arbitrary flange groove
D_Iseal = 15.475;                     % Inner sealing diameter (in)
gland_ID = D_Iseal;
gland_OD = D_Iseal + 2 * (d_cross + land_width);   % Inner + seal + land
both sides
```

```
fprintf('\n=== Axial Seal Gland Dimensions ===\n');
fprintf('O-Ring Cross-Section: %.4f in\n', d_cross);
fprintf('Groove Depth (80%% compression): %.4f in\n', groove_depth);
fprintf('Groove Width (fill limit): %.4f in\n', groove_width);
fprintf('Gland Inner Diameter: %.4f in\n', gland_ID);
fprintf('Gland Outer Diameter: %.4f in\n', gland_OD);
fprintf('Gland Land Width (each side): %.4f in\n', land_width);



fprintf('\n=== Fastener Analysis (Stiffness-Based Load Model, Temp-Adjusted)
===\n');
fprintf('Number of Bolts: %d\n', n_bolts);
fprintf('Bolt Diameter: %.2f in\n', d_bolt);
fprintf('Tensile Stress Area At: %.4f in^2\n', At);
fprintf('Total Required Clamp Load (W_total): %.2f lb\n', W_total);
fprintf('Load Per Bolt (External Only): %.2f lb\n', W_per_bolt);

fprintf('\n--- Safety Factors WITHOUT Preload ---\n');
fprintf('Safety Factor (Yield, Temp-Adjusted): %.2f\n', SFy_no_preload);
fprintf('Safety Factor (Ultimate, Temp-Adjusted): %.2f\n', SFu_no_preload);

fprintf('\n--- Stiffness Model Parameters ---\n');
fprintf('Bolt Stiffness (k_b): %.2e lb/in\n', k_b);
fprintf('Member Stiffness (k_m): %.2e lb/in\n', k_m);
fprintf('Load Fraction Carried by Bolt (C): %.3f\n', C);

fprintf('\n--- Safety Factors WITH Preload + Load Share ---\n');
fprintf('Preload Per Bolt Used: %.2f lb\n', Fi);
fprintf('Total Bolt Load (Fb): %.2f lb\n', Fb);
fprintf('Safety Factor (Yield, Temp-Adjusted): %.2f\n', SFy_with_preload);
fprintf('Safety Factor (Ultimate, Temp-Adjusted): %.2f\n', SFu_with_preload);

fprintf('\n--- Torque Recommendation ---\n');
fprintf('Recommended Assembly Torque: %.1f lb-in\n', Torque);

fprintf('\n--- Flange Separation Check ---\n');
fprintf('Total Clamp Load from Bolts: %.2f lb\n', Fi_total);
fprintf('Load Trying to Separate Joint (C * W): %.2f lb\n', W_separation);
fprintf('Separation Safety Factor: %.2f\n', SF_sep);
```

```
=== Socket Clearance Report ===
Bolt Circle Diameter: 17.000 in
Flange Outer Diameter: 19.250 in
Bolt-to-Bolt Spacing: 2.219 in
Radial Clearance (Bolt to Wall): 1.125 in
---
Socket OD: 1.375 in
Actual Margin Between Bolts: 0.844 in
Actual Radial Margin to Wall: 0.438 in
□ PASS: Socket fits with sufficient margin (≥ 0.2 in).

=== Axial Seal Gland Dimensions ===
```

```
O-Ring Cross-Section: 0.2750 in
Groove Depth (80% compression): 0.2200 in
Groove Width (fill limit): 0.3235 in
Gland Inner Diameter: 15.4750 in
Gland Outer Diameter: 16.1250 in
Gland Land Width (each side): 0.0500 in

=== Fastener Analysis (Stiffness-Based Load Model, Temp-Adjusted) ===
Number of Bolts: 24
Bolt Diameter: 0.75 in
Tensile Stress Area At: 0.3340 in^2
Total Required Clamp Load (W_total): 80676.30 lb
Load Per Bolt (External Only): 17657.24 lb

--- Safety Factors WITHOUT Preload ---
Safety Factor (Yield, Temp-Adjusted): 2.14
Safety Factor (Ultimate, Temp-Adjusted): 3.03

--- Stiffness Model Parameters ---
Bolt Stiffness (k_b): 9.35e+06 lb/in
Member Stiffness (k_m): 1.74e+07 lb/in
Load Fraction Carried by Bolt (C): 0.350

--- Safety Factors WITH Preload + Load Share ---
Preload Per Bolt Used: 17657.24 lb
Total Bolt Load (Fb): 23837.28 lb
Safety Factor (Yield, Temp-Adjusted): 1.59
Safety Factor (Ultimate, Temp-Adjusted): 2.24

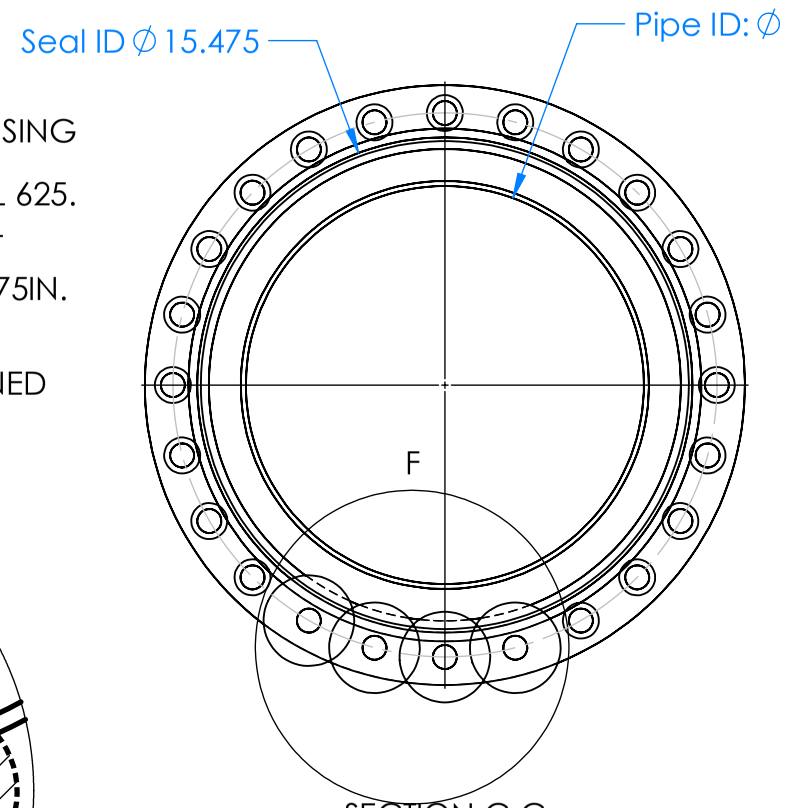
--- Torque Recommendation ---
Recommended Assembly Torque: 2383.7 lb-in

--- Flange Separation Check ---
Total Clamp Load from Bolts: 423773.86 lb
Load Trying to Separate Joint (C * W): 28236.70 lb
Separation Safety Factor: 15.01
```

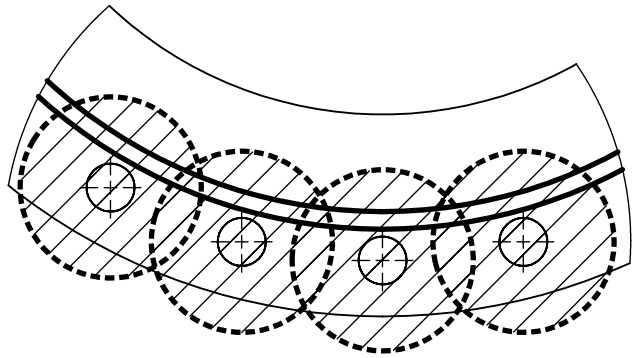
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ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	N/A	Inconel 625 Flange Body/pipe (TOP)	1
2	N/A	Inconel 625 Flange Body/pipe (BOTTOM)	1
3	99904A229	Medium Strength A-286 Steel Serrated Flange Locknut	24
4	91271A849	A-286 12-Point Screw, 3/4"-13 UNC	24
5	O-RING SEAL	16.025 OD Metal O-RING	1

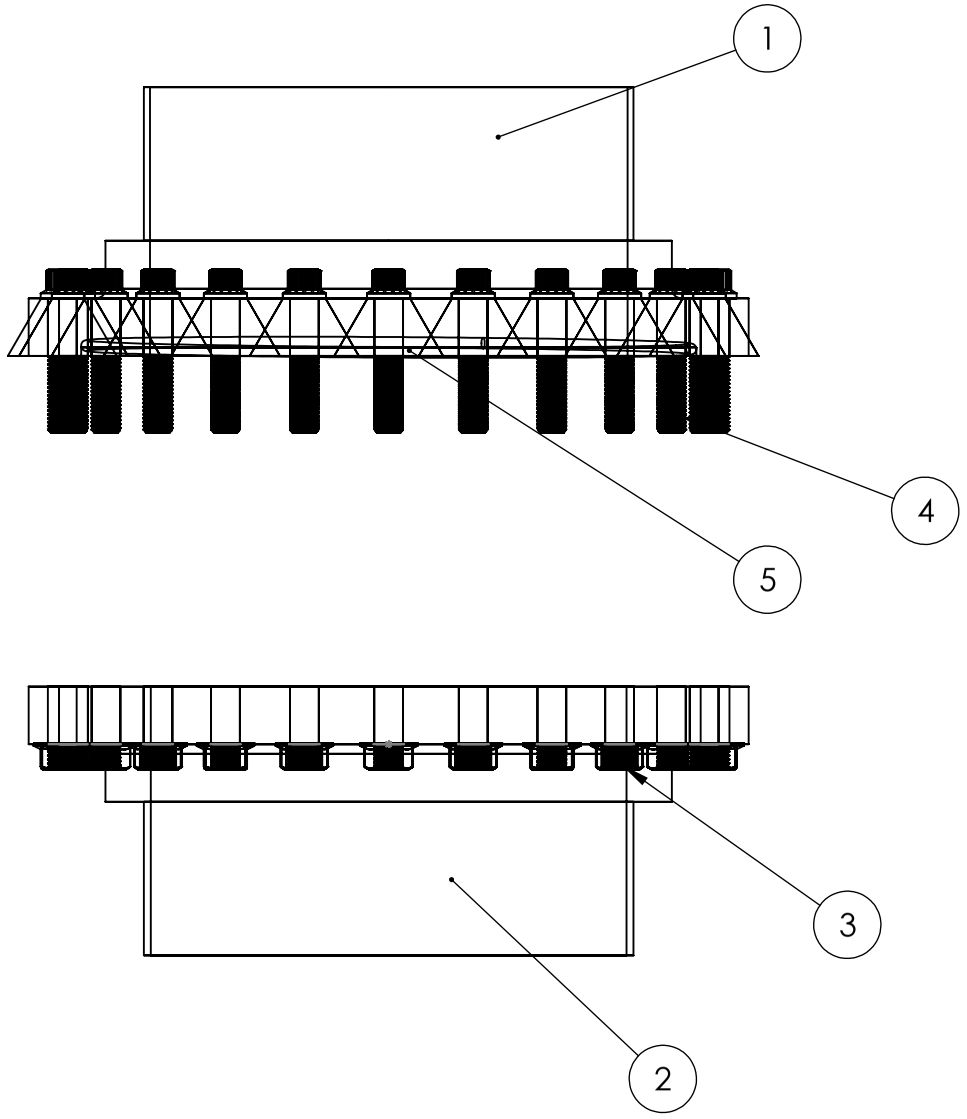
NOTES
UNLESS OTHERWISE SPECIFIED:
1. ALL DIMS. IN INCHES
2. ASSEMBLE BOLTS TO 2383.7 LB-IN TORQUE USING LUBRICATED THREADS (K = 0.18).
3. ALL FLANGE COMPONENTS TO BE INCONEL 625.
4. ALL FASTENERS TO BE A-286 STAINLESS STEEL
5. FLANGE SEAL HAS AN OUTER DIAMETER OF 16.025IN AND AN INNER DIAMETER OF 15.475IN.
6. DESIGN BASED ON MEOP = 400 PSI AND OPERATING TEMPERATURE OF 600 DEG. F
7. DETAIL VIEW F SHOWS SEAL FULLY CONTAINED WITHIN BOLT CONES OF COMPRESSION

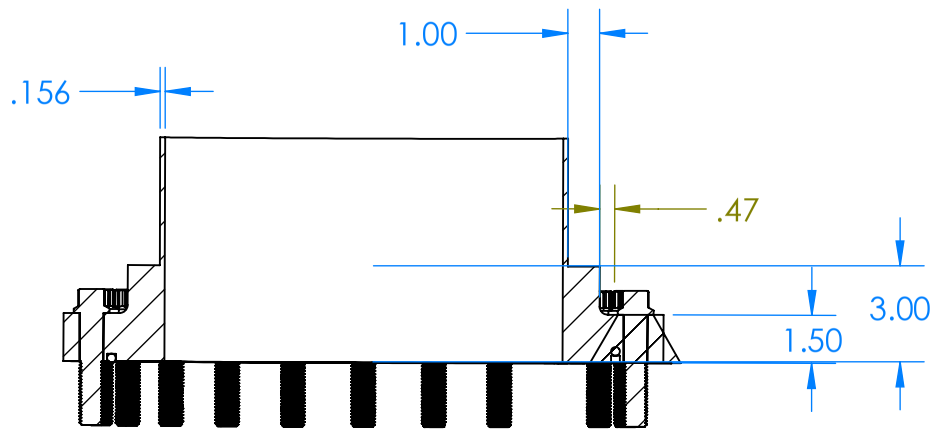


SECTION C-C
SCALE 1 : 6

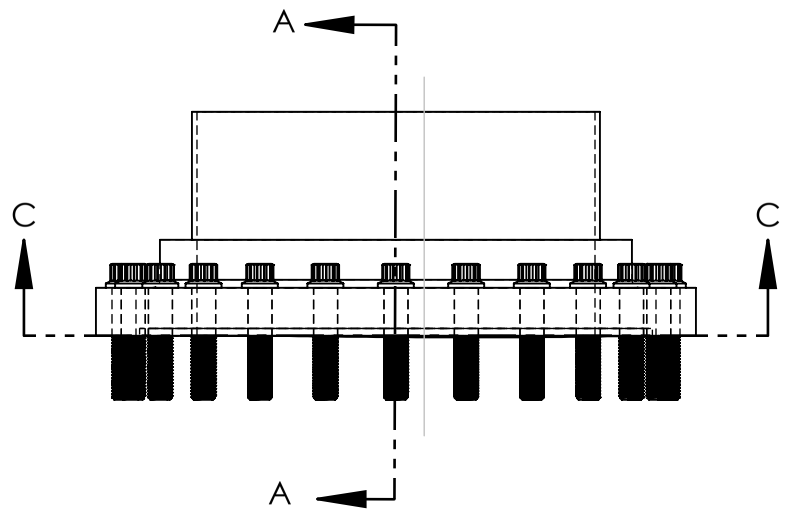


DETAIL F
SCALE 1 : 3





SECTION A-A



Cal Poly Mechanical Engineering ME 329 - SPRING 2025	Lab Section:12 LAB #4	Title: FLANGE DESIGN Date: 5/28/2025	Scale: 1:6	Drwn. By: RYAN TREVENA AND GABRIEL CHOI

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